

Virtual Animal Embodiment for Actor Training

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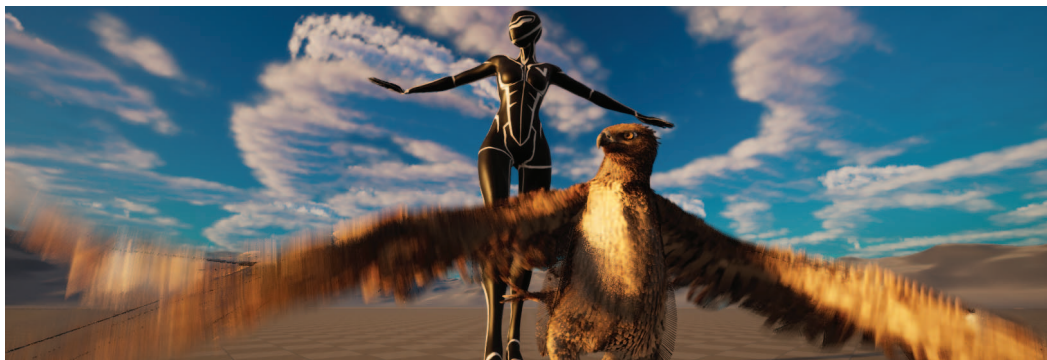


Figure 1: A screenshot of an actor performance. Shown is the real-time virtual avatar and the motion applied to a high-fidelity eagle model. In the final system, the actor will view only the eagle avatar with their motions applied.

ABSTRACT

In this work-in-progress paper, we discuss the development of a real-time system allowing actors to drive the motion of an animal avatar, our test case being a photorealistic eagle. The aim of the work is to enhance actor training for studies related to the connection between the mind and body, imagination and action. In our proposed perceptual experiments, we plan to assess the quality of experience for the actors and also if there is a performance improvement due to the embodiment.

Index Terms: Human-centered computing—Interaction design—Epirical studies in interaction design; Human-centered computing—Interaction paradigms—Virtual reality

1 INTRODUCTION

Students of acting often engage in studies aimed to improve expressive use of the body and to learn a precise and articulate physical vocabulary. One such exercise is an Animal Study where students are asked to choose an animal and observe it using Laban Movement Analysis (LMA) principles and to apply the work to character transformation including the ability to have spontaneous thoughts and instincts of the chosen animal. Students aim to foster a seamless connection between mind and body, imagination and action and are asked to perform with freedom, finesse, variety and precision between movement qualities. Additionally, to develop spatial awareness and to practice posture and grounding.

In this work, we propose the use of technology, such as Virtual Reality, to allow actors to enhance their studies by pushing the transformation from human to animal further using motion capture and virtual environments. We aim to provide them with high-quality

animal avatars which they can animate with their body movements in real-time with high-quality full-body optical motion capture. This will give them a better sense of the animal's size in relation to its environment, as well as a greater understanding of the constraints of body movement. We aim to investigate if such an approach could enhance the embodiment experience, pushing the actor to transform further and create a more convincing performance. In summary, our long-term objective is to provide actors with a training tool that enables them to embody virtual animal characters and visualize their performances in real-time within a virtual environment, either on screen or through a virtual reality headset. We aim to develop a system capable of generating realistic animal motions while seamlessly aligning with actor movement performances. Finding a balance between motion realism and performance coherence is crucial for such system. To achieve this, we plan to incorporate feedback from actors and trainers throughout the development process. The culmination of this effort will be a user study designed to evaluate the advantages of employing such a technique. In this paper, we describe the design of our preliminary animal embodiment system for actor training, the current set-up implemented and tested on an eagle model, the plan for future investigations to improve the retargeting techniques, and the final user evaluation involving actors, to study the effects of virtual technologies for embodiment and immersion. This work is organized as follows: first, in Section 2 we summarize a set of relevant related works, in Section 3 we introduce the proposed system, describe the preliminary implementation, and finally, in Section 4 we discuss current limitations and future solutions.

2 RELATED WORK

2.1 Technology-enhanced acting

Previous works have investigated the use of Virtual Reality (VR) and motion capture to improve acting performances. Kammerlander et al. [12] used motion-captured performances in VR to bring differently proportioned virtual characters into a shared collaborative virtual environment. The embodiment enhanced actor's feelings of body ownership and immersion. Additionally, this changed the performances, allowing for better animation output. In other work,

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Table 1: Previous studies on Animal Embodiment in VR.

Experiment	Animal	Feedback	Measures
Egan et al. [6]	dog	visual	Body Ownership
Vargas et al. [24]	dog	visual, haptic, audio	Body Ownership, Empathy
Ahn et al. [1]	shorthorn cow, coral	visual, haptic, audio	Body transfer, Spatial presence, Inclusion of Nature in Self
Krehov et al. [15]	scorpion, rhino, bird	visuo-motor	Immersive Tendencies, Intrinsic Motivation Inventory, Player Experience of Need Satisfaction, Presence, Simulator Sickness
Krehov et al. [16]	bat, tiger, spider	visuo-motor	Acceptance, control, change, fascination, ease of control, fatigue
Li et al. [17]	cat	visuo-motor, haptic	Body ownership, Self-Illusion
Oyanagi et al. [18]	bird	visuo-motor, audio	Body ownership, change, agency, feature
Oyanagi et al. [19]	bird	visuo-motor	Sense of flight, agency, ownership, immersion, reality, self-location, transformation, anxiety about falling, fear of heights
Pimentel et al. [20]	loggerhead turtle	visuo-motor, visuo-tactile	Environmental concern, environmentalism, spatial presence, copresence, donation intention, nature in self, threat perception, and conservation behavioural intentions
Sierra Rativa et al. [23]	beaver	visual, haptic	Dispositional empathy, self-reported situational empathy, immersion, VR sickness symptoms, self-reported emotions, self-reported animal conservation tendency, perceived pain, embodiment

Bouville et al. [4] proposed a system for rehearsals in VR, where actors could familiarize themselves with their virtual surroundings and act out a scene with a virtual tiger. The study showed a preference for VR training over classical training as actors had a better vision of the scene and able to try out more interactions with the virtual tiger. Kade et al. [11] found that mixed reality virtual environments are useful as acting support for motion-capture actors who typically have to imagine the virtual scenes they are acting in. More recently, Cannavo et al. [5] proposed an Augmented Reality (AR) system allowing the virtual characters to be overlaid onto the motion capture actor's bodies. They found positive effects of their system, in particular in helping actors to position themselves and direct their gaze while shooting a scene. Following a similar approach to previous studies, we intend to assess the impact of our tool on both performances and perceptual metrics.

2.2 Animal Embodiment

Animal embodiment comes with challenges that must be considered to elicit an effective Sense of Embodiment (SoE) (the feeling of being inside, having, and controlling their virtual body [13]). Krehov et al. [16] investigated animal embodiment with rhino, scorpion, and bird avatars, and found high player enjoyment in escape room games. Interestingly, they found that the animal's additional body parts and skills enhanced rather than hindered players SoE, unlocking the hidden potential of animal avatars. Providing rich multi-sensory feedback has also been shown to increase the SoE towards virtual animals [1, 20, 23, 24]. Participants in various studies have been exposed to visuo-motor, visuo-tactile, and audio-visual stimulation in an effort to increase their SoE. For example, Vargas et al. [24] investigated the impact of auditory and visuo-tactile feedback by immersing participants in the body of a dog and providing them with synchronous tactile stimulation and auditory feedback. Additionally, Ahn et al. [1] exposed participants of their experiments to visual, haptic and auditory feedback while embodied in a shorthorn cow and a coral on a rocky reef. They found increased SoE, and presence as a result of the rich multi-sensory feedback. Our study aims to use visuo-motor feedback initially, but we will also investigate auditory feedback in the form of congruent animal sounds. Tactile feedback may also be investigated in order to enhance the feeling of owning dissimilar body parts like wings.

The Proteus effect, similar to self-illusion, postulates that a user's behaviour, attitude, and mental state are affected by the avatar's appearance [25]. Oyanagi et al. have investigated the Proteus effect by embodying participants into a bird avatar [18, 19]. They found that a bird avatar can significantly improve the psychological experience of flying, and reduced participants' fear of heights. Similarly, Li et al. [17] conducted an experiment in which participants played the role of a cat and discovered that a user's high level of self-illusion generally contributes to a greater connection to the vir-

tual role, which is promising for our approach. We postulate that the Proteus effect may further enhance the actors transformation into their chosen animals. See table 1 for a summary of animal embodiment studies, including the types of animal avatar used in the study, sensory feedback, and the measures included in the experiments.

2.3 Human-animal mapping

The success of an animal embodiment system relies on the real-time mapping of the actor's motions to the target animal. Extreme morphological differences and the complexities of different animals' motions can make the construction of this mapping challenging. There can be a trade-off between actor's agency levels and the realism of the output motions of the target animal. Many existing animal embodiment works rely on inverse kinematics (IK) for this mapping [1, 15, 19, 20, 23]. The use of IK results in high levels of agency but potentially low animation realism for the target animal.

Alternative methods to IK have also been explored for mapping human motion to animal motion. Many of these methods are more akin to puppetry systems in which the user controls the target animal, for example, using gestures (e.g. [21]) or by performing a limited set of pre-determined motions (e.g. [22]). For example, some works have investigated mapping the motion of the actor's hand to the target animal [10, 14]. Seol et al. [22] present an animal puppetry system in which the final output animal pose is a blend between a pose resulting from learned linear mappings between input human features and target animal features and a pose determined via a frame-by-frame correspondence constructed from reference human motions and target animal animations. Rhodin et al. [21] estimate wave properties (amplitude, frequency and phase) from input user wave gestures in real-time and use these to animate a target character (for example, a quadruped or a caterpillar) via a parametric motion graph ([8]) in which the example animations have been annotated with amplitude, frequency and phase parameters. In contrast to the puppetry systems above, more recent work has used deep-learning to present mappings which allow the actor to move freely and naturally with their motion relatively unconstrained [3, 6]. Boehs and Viera [3] use neural networks to appropriately select a frame to play back from pre-generated animal animations based on the actor's motion. Egan et al. [6] use a Mixture of Experts [26] neural network to synthesise realistic quadruped motion conditioned on the actor's global trajectory and the motion of their legs, while in other work uses neural networks to add subtleties and correct errors to produce an enhanced animal animation [7].

In our work, we aim to explore these recent approaches to define a flexible methodology that can be adapted to various animal morphologies, with a specific focus on capturing and portraying actors' full-body performances.

3 SYSTEM

The goal of this work is to capture and transfer human performers' motion onto virtual animal characters in real time, in order to produce plausible visual feedback which could potentially support actors during training and enhance their performances.

With this intent, we identify a set of general technical requirements: *i)* the system should adapt to a wide variety of body structures and proportions to allow the embodiment of various animal species. *ii)* The system should preserve user agency and control over the animal movements, while, at the same time, *iii)* guarantee the plausibility of the generated motion. The appropriate integration of these technical requirements is what we will explore in the planned user study. Our exploration starts from a technical implementation adapted for a single introductory eagle-embodiment scenario. In this paragraph, we detail the proposed preliminary set-up, commenting on the current limitations and the expected future implementation. We aim for high photorealism of the character as it has been shown to enhance appeal [9] place illusion [27] and social presence [28] which may prove important for the actor embodiment experience. Therefore, we selected a high-end rigged eagle character with realistic feathers created by Weta Digital and freely available in the Unreal Engine Marketplace¹. In our preliminary configuration, the system

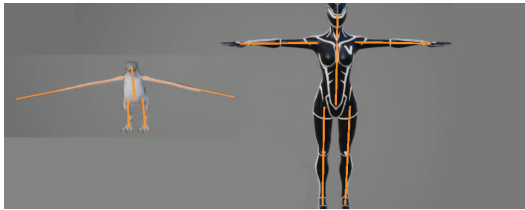


Figure 2: Comparison of the two rigged models employed in our preliminary system. Retargeted limbs – bone chains – are highlighted in orange.

captures performer's movements using a Vicon optical motion capture system consisting of 13 Vantage and 8 MX T40 cameras and 52 markers. The captured motion is streamed directly into Unreal Engine 5.3, where the human motion is reconstructed and retargeted onto the Eagle character. The generated visual feedback is then displayed either on an LCD screen or on a VR head mounted display. The latter displays both an embodied first-person-view avatar and a mirrored image, while the first only display the mirrored one. In this preliminary phase, we rely on an existing retargeting tool: Unreal Engine's IK Rig Retargeter, which animates the selected character proportionally to the kinetic properties of the source human performance. The tool operates on the correspondence between bone chain structures from the source skeletal structure to the target one, like the spine, neck, arms, and legs. In contrast with traditional retargeting implementations, this approach attempts to transfer motions between slightly dissimilar and disproportional skeletal configurations, for instance, when joint numbers differ or when limbs have different lengths. In fact, the movement from a source chain is transferred to the corresponding target chain, where it is adapted to the proportions of the target body. Our implementation adapts these retargeting properties for the eagle character embodiment (see Fig. 2). Intuitively, we associate the motion of the human legs, spine, and head with that of the corresponding body parts on the eagle and the motion of the human arms with that of the eagle's wings. A relevant advantage of the proposed implementation is the employment of IK which helps prevent excessive anthropomorphism of the animal motion. Instead of transferring rotations from joint to joint, the system only sets the

extremities' properties of the chain and adjusts the rotations of the in-between joints to fit the generated configuration. An example of this is illustrated in Fig. 3 where the retargeting preserves the eagle's leg articulation while walking. However, this discrepancy might affect the performer's perceived embodiment and sense of agency over the eagle model. The planned user study aims to investigate these effects, exploring the SoE and agency of the participant while evaluating the produced performances.



Figure 3: Detail of the legs of the source (left) and target (right) animation. The employed retargeting system preserves the original leg configuration on the eagle model (highlighted in blue) while the motion is generated from the chain correspondence with the leg of the source model (highlighted in red).

4 DISCUSSION AND FUTURE WORK

The described preliminary exploration allows us to identify drawbacks and schedule the next steps of our study. As described in Section 2, the process of embodying humans in animal characters presents various challenges. Those include the balance between motion realism and user agency, especially considering motions that are non-reproducible by a human due to morphological differences with the variety of animal creatures. In this paper, we relied on a tool designed to transfer movements between characters with similar morphology and comparable motion dynamics, such as humanoids. The presented examples demonstrate how to extend the same approach to generate motion for a distinct morphology, such as the eagle, which exhibits different movement traits. More work needs to be done to include highly distinct morphologies, such as spiders, or to produce a higher variety of actions, like flying for birds. In the coming months, we plan to overcome these challenges by improving the retargeting system. Firstly, we aim to develop an approach to identify and reproduce various actions. This may require specific tuning for each creature. However, methods such as example-based calibration and finite-state machines should help reduce manual effort. Such process involves an initial phase where the performer reproduces a set of animal movements to calibrate the retargeting parameters, and later identifies and triggers the appropriate actions at runtime. Secondly, additional improvements will focus on the appropriate generation of animal movement. To achieve this, we plan to explore state-of-the-art techniques both to enhance the realism of the generated animation and to respect the anatomical body constraints. We intend to delve into two distinct approaches: either employ physical constraints to improve the generation of plausible motions [2], or learn it from animal data [6]. In fact, after identifying actions and motion parameters, the system requires the reconstruction of the target animal motion. In the presented approach, we rely solely on the kinematic properties of bone chains. The incorporation of physics parameters and the integration of learned features from animal motion data not only enhance the richness of, but also contribute to a more accurate foundation for motion retargeting. As the use case of the system is to help with actor training, the implementation will be an iterative process, incorporating the feedback of actors along the

¹<https://docs.unrealengine.com/4.27/en-US/Resources/Showcases/Meerkat/>

way. This phase involves actors testing the system, participating in interviews and responding to questionnaires. Additionally, acting teachers will offer further insight on their performances. This encompasses the evaluation of elements like embodiment, agency, personal feedback, and the professor's judgment. The collected data will also contribute to understanding the effects of motion styles and human morphologies on the generated animal motion. Ultimately, we aim to conduct a comprehensive user study to compare the mentioned elements, in order to evaluate the proposed technology in comparison with other techniques and with pure performance, meaning with no technological support.

The refinement of animal embodiment for actor training represents a crucial advancement in integrating interactive technologies into performing arts. Investigating such an approach will strengthen our understanding of motion properties, which are essential for a seamless transition between human and animal movement. This study not only contributes to actor training but also extends its benefits to a broader range of fields, including film production, immersive interactive experiences such as video games, and social virtual reality. Finally, by gathering feedback from actors, we aim to unravel the complexity involved in this transfer, contributing significantly to our overall comprehension of motion.

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